

# Can policy gradient replicate the exact soccer solver?

REINFORCE, A2C and PPO, trained by self-play and by fictitious play, scored against the exact solution of the discounted 100-step soccer game. Every number below is read from the experiment files in the repository.

## Summary

- Six learners, REINFORCE, A2C and PPO each trained by self-play and by fictitious play, on the discounted 100-step soccer game, scored against the exact solution.
- Their exploitability ranges from 0.76 to 0.95; the exact solution is 0.
- Fictitious-play training wins more often against a random player; self-play training ties the exact equilibrium more often.
- The two players' policies are not mirror images (mirror gap 0.44 to 0.81; 0 for the exact solution).

Results are for 3 seeds at one untuned budget. Every assumption is listed in section 9.

## 1. What was done

These choices fix the setup used throughout.

| topic              | what was done                                                                                                                                                                                 | why                                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Environment        | The deterministic A10 board is the main environment; the random move-order board is kept as an earlier comparison.                                                                            | The deterministic board has no mixed-equilibrium stages, so the comparison with the exact solution is clean. |
| Objective          | Discount 0.9 over 100 steps, a tie at the end, and the remaining step count as a network input.                                                                                               | Policy gradient can only approximate a finite-horizon discounted reward.                                     |
| Fictitious play    | Each player is trained by policy gradient against the average of the opponent's past policies. Matrix-level fictitious play is kept only as a side check (section 7).                         | Policy gradient does the solving; no game is solved explicitly.                                              |
| Win rate           | 1,000 repeated games from the kickoff against a random player, the exact Nash policy and the exact best response; wins, ties and losses are all counted.                                      | Repeated play and counting wins.                                                                             |
| Continuous actions | The dog game uses an angle-radius policy trained directly by policy gradient (section 8, provisional). The bisection, finite-difference and quadratic best-response methods exist separately. | Policy gradient adapts to continuous actions directly; the best-response search methods belong to DQN.       |

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## 2. Setup

The environment is the deterministic A10 soccer game on a 7×5 board with a three-cell goal; it has no mixed-strategy stage games, which keeps the comparison with the exact solution clean. The objective is **discounted** ( $\gamma = 0.9$ ) over a **100-step horizon**, a tie if nobody scores, and the network receives the **remaining step count** as an input. The reference is the exact discounted backward induction over the same 100 steps ( `soccer_nash/finite_horizon.py` ), whose equilibrium has exploitability 0 and ties itself every game.

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## 3. The learners

| algorithm | update                                                |
|-----------|-------------------------------------------------------|
| REINFORCE | plain discounted returns, no baseline                 |
| A2C       | 10-step bootstrapped advantage from a learned critic  |
| PPO       | GAE ( $\lambda = 0.95$ ), clipped ratio 0.2, 4 epochs |

**Self-play** trains both current networks against each other. **Fictitious play** is how policy gradient solves the game here, with no matrix solving: each player keeps improving its own network by policy gradient against the *empirical average* of the opponent's past policies (frozen snapshots, one drawn per episode), and reports the average of its own snapshots. All learners use learning rate  $10^{-3}$ , 64 episodes per batch, 2,000 iterations, an entropy bonus of 0.01, and a 64×64 network; none was tuned.

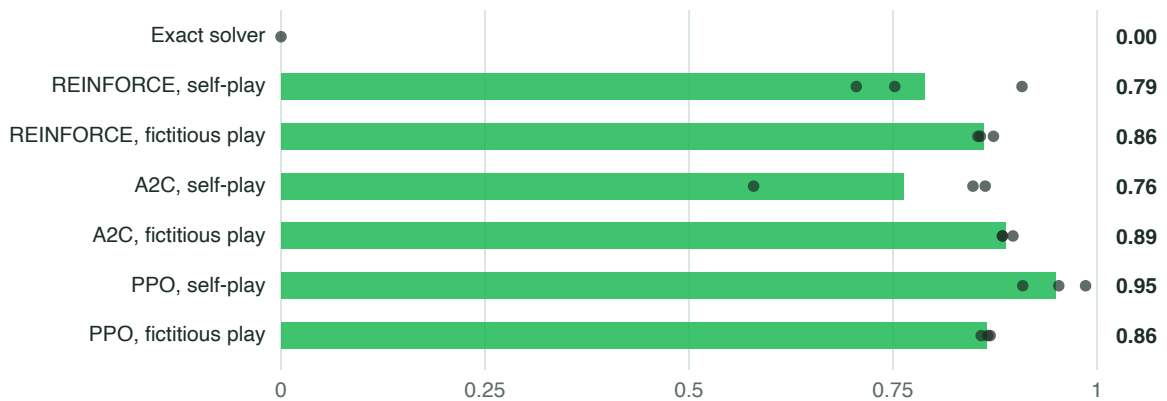
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## 4. How they are scored

- **Exploitability**: the discounted value both exact best responses earn from the kickoff, summed. Zero exactly at an equilibrium.
- **Wins, ties and losses**: counts over 1,000 repeated games from the kickoff, against a random player, the exact Nash policy and the exact best response.
- **Mirror gap**: how far the two players' policies are from being mirror images (flip the board, swap the players), averaged over all states at five points in the game.

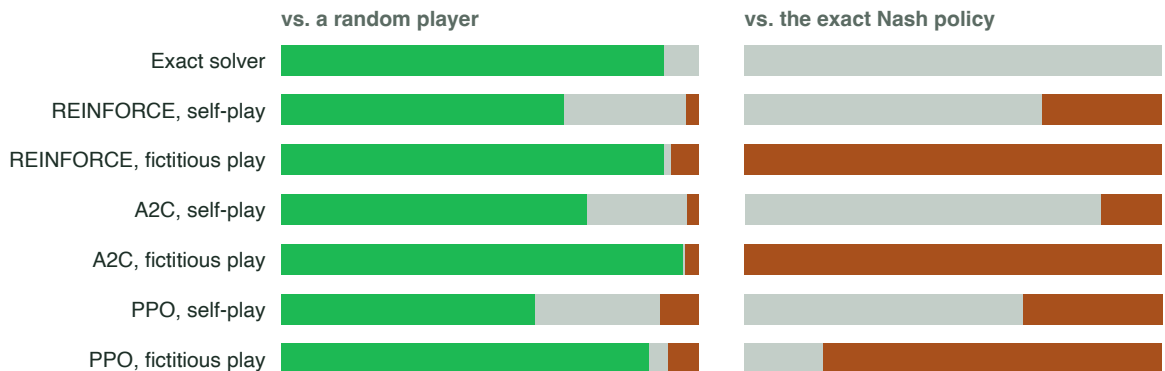
## 5. Results

| learner                          | exploitability<br>(mean $\pm$ sd) | W/T/L vs.<br>random | vs. exact Nash | vs. best<br>response | mirror<br>gap |
|----------------------------------|-----------------------------------|---------------------|----------------|----------------------|---------------|
| Exact solver                     | 0.000 $\pm$ 0.000                 | 0.92/0.08/0.00      | 0.00/1.00/0.00 | 0.00/1.00/0.00       | 0.00          |
| REINFORCE,<br>self-play          | 0.788 $\pm$ 0.106                 | 0.68/0.29/0.03      | 0.00/0.71/0.29 | 0.00/0.00/1.00       | 0.66          |
| REINFORCE,<br>fictitious<br>play | 0.861 $\pm$ 0.010                 | 0.92/0.02/0.07      | 0.00/0.00/1.00 | 0.02/0.00/0.98       | 0.44          |
| A2C, self-<br>play               | 0.763 $\pm$ 0.160                 | 0.73/0.24/0.03      | 0.00/0.85/0.15 | 0.02/0.00/0.98       | 0.53          |
| A2C,<br>fictitious<br>play       | 0.888 $\pm$ 0.008                 | 0.96/0.00/0.03      | 0.00/0.00/1.00 | 0.01/0.00/0.99       | 0.44          |
| PP0, self-<br>play               | 0.949 $\pm$ 0.039                 | 0.61/0.30/0.09      | 0.00/0.67/0.33 | 0.00/0.00/1.00       | 0.81          |
| PP0,<br>fictitious<br>play       | 0.864 $\pm$ 0.006                 | 0.88/0.05/0.07      | 0.00/0.19/0.81 | 0.00/0.00/1.00       | 0.48          |



Exploitability by learner: bar = mean, dots = individual seeds.

win tie loss



Two patterns repeat. Fictitious-play training beats a random player more often (0.88 to 0.96) but loses almost every game to the exact equilibrium. Self-play training wins less often against random and ties

the equilibrium more often: it plays more cautiously. Neither is close to equilibrium play, and every learner loses almost every game to the exact best response.

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## Does more training help?

Self-play with **8,000** iterations instead of 2,000 (2 seeds per learner; every other setting unchanged). The fictitious-play runs were too slow to repeat at this length.

| learner                 | exploitability,<br>2,000 it. (3-seed<br>mean) | exploitability,<br>8,000 it. (each<br>seed) | W/T/L vs.<br>random (8,000<br>it.) | W/T/L vs. exact<br>Nash (8,000 it.) |
|-------------------------|-----------------------------------------------|---------------------------------------------|------------------------------------|-------------------------------------|
| REINFORCE,<br>self-play | 0.788                                         | 0.49 / 0.91                                 | 0.91/0.08/0.02                     | 0.00/0.38/0.62                      |
| A2C, self-<br>play      | 0.763                                         | 0.44 / 0.50                                 | 0.79/0.20/0.01                     | 0.00/0.79/0.21                      |
| PPO, self-<br>play      | 0.949                                         | 0.91 / 0.96                                 | 0.97/0.01/0.02                     | 0.00/0.00/1.00                      |

REINFORCE ends clearly lower on 1 of 2 seeds; A2C ends clearly lower on 2 of 2 seeds; PPO ends clearly lower on 0 of 2 seeds; the best single run reaches 0.44, still far from 0. With two seeds per learner this is a trend, not a result.

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## Fictitious play with best-response phases

A stricter version of fictitious play: in each round each player runs 100 (or 300) policy-gradient iterations to approximate a best response to the average of the opponent's earlier best responses, then adds that policy to its history. The reported policy is the per-state average of the best responses. Exploitability after each checkpoint round (1 seed per run, 64 episodes per iteration):

| run                                            | round<br>5 | round<br>10 | round<br>15 | round<br>20 | round<br>25 | round<br>30 |
|------------------------------------------------|------------|-------------|-------------|-------------|-------------|-------------|
| A2C, 100 iterations per best<br>response       | 0.85       | 0.88        | 0.89        | 0.90        | 0.91        | 0.92        |
| A2C, 300 iterations per best<br>response       | 0.89       | 0.92        | 0.93        |             |             |             |
| PPO, 100 iterations per best<br>response       | 0.89       | 0.92        | 0.90        | 0.90        | 0.91        | 0.91        |
| REINFORCE, 100 iterations per<br>best response | 0.85       | 0.85        | 0.87        | 0.86        | 0.87        | 0.87        |

In every run the exploitability at the last checkpoint is higher than at the first; none comes close to 0.

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## 6. Earlier experiments

Earlier, the same algorithms were run on the stationary discounted game (no step count), from the kickoff or from random starting states.

| setup                                      | algorithm | exploitability | mean equilibrium regret | action agreement |
|--------------------------------------------|-----------|----------------|-------------------------|------------------|
| A10 board, stationary, kickoff starts      | REINFORCE | 0.83           | 0.124                   | 0.28             |
| A10 board, stationary, kickoff starts      | A2C       | 0.83           | 0.122                   | 0.29             |
| A10 board, stationary, kickoff starts      | PP0       | 0.87           | 0.167                   | 0.25             |
| random board, stationary, kickoff starts   | REINFORCE | 0.73           | 0.056                   | 0.44             |
| random board, stationary, kickoff starts   | A2C       | 0.78           | 0.058                   | 0.45             |
| random board, stationary, kickoff starts   | PP0       | 0.85           | 0.108                   | 0.32             |
| random board, stationary, exploring starts | REINFORCE | 0.80           | 0.027                   | 0.71             |
| random board, stationary, exploring starts | A2C       | 0.82           | 0.030                   | 0.70             |
| random board, stationary, exploring starts | PP0       | 0.91           | 0.072                   | 0.51             |

Exploring starts raise per-state accuracy (action agreement) but not play from the kickoff.

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## 7. Fictitious play on the exact stage games

As a side check, fictitious play was also run on the exact 4×4 stage matrices of the random move-order board (2,380 states, 94 with no pure saddle). On rock-paper-scissors best-response dynamics cycle forever while fictitious play settles at 1/3 each; on the soccer stage games it recovers the exact values.

| rounds | mean stage-game gap | max stage-game gap | max value error |
|--------|---------------------|--------------------|-----------------|
| 100    | 0.00233             | 0.02020            | 0.01010         |
| 1000   | 0.00026             | 0.00296            | 0.00133         |
| 10000  | 0.00004             | 0.00094            | 0.00037         |

This is matrix-level fictitious play. Here policy gradient itself does the solving, which is what section 5 tests.

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## 8. Continuous actions: the dog game (provisional)

Policy gradient is applied to continuous actions directly: a network outputs an angle (von Mises) and a radius (scaled Beta) and is trained by self-play PPO. The dog-and-sheep game itself is **my placeholder**, since no definition exists yet, so these numbers say nothing about equilibrium.

| matchup                            | capture rate | mean capture step |
|------------------------------------|--------------|-------------------|
| random dog vs random sheep         | 0.00         | 96                |
| trained dog vs random sheep        | 0.26         | 41                |
| random dog vs trained sheep        | 0.00         | –                 |
| trained dog vs trained sheep       | 1.00         | 30                |
| straight-line dog vs trained sheep | 1.00         | 19                |

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## 9. Assumptions and open questions

All 27 assumptions are listed, each marked as quoted, answered, mine or invented, in `docs/solver_assumptions.md`. Still open: the dog-game rules, the polar-policy distribution, what “check symmetry” should mean, which states to evaluate on, how degenerate equilibria should be compared, and two page-3 items (the “R game” closed form and “# time f is activated”).

Reproduce: `python scripts/pg_finite.py --seeds 3`, then `make pg-pdf`.